MCS-332	NETWORKING PROTOCOLS	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To understand the basic concepts of data communication, layered model, protocols and interworking between computer networks and switching components in telecommunication systems.
CO2	Discuss the nature, uses and implications of internet technology.
CO3	To understand the functioning of Frame Relay, ATM
CO4	An overview of security issues related to data communication in networks.
CO5	Know about different application layer protocols

MODULE I

Networks and Services, Approaches to Network Design, The OSI Reference Model; Overview of TCP/IP Architecture, Application Protocols and TCP/IP Utilities, Internet Architecture Interconnection through IP Routers, Internet Protocol (IP), User datagram protocol (UDP).

MODULE II

Routing Cores - peers Algorithms Autonomous Systems Exterior Gateway Protocol Multicast Address. Internet Group **Management** Protocol (IGMP) and Implementation. TCP/IP over ATM networks: ATM cell Transport , Adaptation Layer, IP Address Building in an ATM network Logical IP subnet Concept ATM-ARP packet format. Domain name **system** , Remote Login (Telnet, Rlogin) File Transfer and Access (FTP, TFTP, NFS), Electronic mail (SMTP, MIME) Internet **Management** (SNMP, SNMPV2) Internet Security and Firewall Design Post Office Protocol (POP) Network News Transfer Protocol (NNTP).

MODULE III

TCP/IP over view- The Transport Layer: TCP and UDP. Elementary TCP Sockets. TCP Client-Server Example. I/O Multiplexing: The select and poll Functions. Socket Options. Elementary UDP Sockets. Elementary Name and Address Conversions.

The Client Server Model and Software Design, Concurrent Processing in Client-Server Software, Iterative, Connectionless Servers (UDP), Iterative, Connection-Oriented Servers (TCP), Concurrent, Connection-Oriented Servers (TCP). Single-Process, Concurrent Servers (TCP). Multiprotocol Servers (TCP, UDP), Multiservice Servers (TCP, UDP). Uniform, Efficient **Management** of server. Concurrency in clients. TCP/IP Architecture, The Internet Protocol, Limitations of IPv4 and Introduction to IPv6, User Datagram Protocol, Transmission Control Protocol, DHCP, Introduction to Internet Routing Protocols

References:

1. *A. Leon-Garcia, Indra Widjaja, "Communication Networks", Tata McGraw Hill, 2000*
2. *William Stallings, "Data and Computer Communications", Pearson Education, 7 th Edition.*
3. *Andrew S. Tanenbaum, "Computer Networks", Prentice Hall India, 4th Edition, 2003*
4. *W.Richard Stevens: TCP/IP Illustrated vol 1: The Protocols, Pearson Edun. Asia, 2000.*
5. *Douglas Comer: Internetworking with TCP/IP vol.1: Principles, Protocols and Architecture, Prentice Hall, 4th edition, 2000*

Course Outcome:

CO1	Describe the basis and structure of an abstract layered protocol model
CO2	Independently understand basic computer network technology.
CO3	Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.
CO4	Identify the different types of network devices and their functions within a network
CO5	Familiarity with the basic protocols of computer networks, and how they can be used to assist in network design and implementation